

Chapter 6

Total Probability and Bayes' Theorem

Lecture 6 | Devore 9e: 2.4–2.5

Last time: conditional probability, the multiplication rule, and independence of two events. We ended by defining pairwise and mutual independence and claiming they differ. Let us settle that first.

6.1 Pairwise does not imply mutual

Example 6.1 (Bernstein's tetrahedron). A regular tetrahedron has four faces, painted

red, white, black, red + white + black (all three).

Roll it and look at the face it lands on. Let R , W , K be the events that the resting face contains red, white, black respectively. Each color appears on exactly two of the four faces, so

$$\mathbb{P}(R) = \mathbb{P}(W) = \mathbb{P}(K) = \frac{2}{4} = \frac{1}{2}.$$

Two colors appear together on exactly one face (the tricolor one), so

$$\mathbb{P}(R \cap W) = \mathbb{P}(R \cap K) = \mathbb{P}(W \cap K) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}.$$

So the three events are **pairwise independent**. But

$$\mathbb{P}(R \cap W \cap K) = \frac{1}{4} \neq \frac{1}{8} = \mathbb{P}(R)\mathbb{P}(W)\mathbb{P}(K),$$

so they are **not mutually independent**.

The moral: pairwise independence lets you multiply two at a time and nothing more. Whenever a problem says “independent” about three or more events, it means mutually independent.

6.2 The law of total probability

Theorem 6.2 (Law of total probability). *Let $\{A_i\}_{i \geq 1}$ be a partition of S with $\mathbb{P}(A_i) > 0$ for*

every i . Then for any event B ,

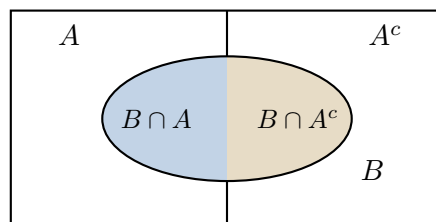
$$\mathbb{P}(B) = \sum_i \mathbb{P}(A_i) \mathbb{P}(B \mid A_i).$$

In the simplest case ($A_1 = A$, $A_2 = A^c$): $\mathbb{P}(B) = \mathbb{P}(A)\mathbb{P}(B \mid A) + \mathbb{P}(A^c)\mathbb{P}(B \mid A^c)$.

Proof. Since $\bigcup_i A_i = S$,

$$\mathbb{P}(B) = \mathbb{P}\left(B \cap \bigcup_i A_i\right) = \mathbb{P}\left(\bigcup_i (B \cap A_i)\right) = \sum_i \mathbb{P}(B \cap A_i) = \sum_i \mathbb{P}(A_i) \mathbb{P}(B \mid A_i),$$

where the third equality is countable additivity: the sets $B \cap A_i$ are pairwise disjoint because the A_i are. $\square$



Remark 6.3. This is the workhorse. Whenever a problem has a hidden “which case are we in?” — which machine made the part, which route the person took, which box the coin came from — condition on the cases and add up.

Example 6.4 (Lost umbrella). You lost your umbrella in one of three places, with prior probabilities

$$\mathbb{P}(A_1) = 0.50 \text{ (library)}, \quad \mathbb{P}(A_2) = 0.30 \text{ (classroom)}, \quad \mathbb{P}(A_3) = 0.20 \text{ (bus)},$$

and if it is there, the chance you actually recover it is

$$\mathbb{P}(B \mid A_1) = 0.80, \quad \mathbb{P}(B \mid A_2) = 0.60, \quad \mathbb{P}(B \mid A_3) = 0.05,$$

where $B = \{\text{umbrella is found}\}$. Then

$$\mathbb{P}(B) = 0.50(0.80) + 0.30(0.60) + 0.20(0.05) = 0.40 + 0.18 + 0.01 = 0.59.$$

6.2.1 Conditioning on the first step

One use of “condition on the cases” deserves its own name. When a game can go on for any number of rounds, do not try to add up all the ways it can end. Condition on what happens in the *first* round instead. If the game then either ends or returns to a position you have already named, the law of total probability gives an *equation* for the unknown probability — or a small system of equations, one unknown per position. Independence is what makes “returns to the same position” legitimate: later rounds do not remember earlier ones. This is called **first-step analysis**.

Example 6.5 (First to roll a six). Alice and Bob take turns rolling a fair die, Alice first. Whoever rolls the first six wins. What is the probability that Alice wins?

Let p be that probability, and condition on the first two rolls. Alice wins at once with probability $\frac{1}{6}$. With probability $\frac{5}{6} \cdot \frac{5}{6} = \frac{25}{36}$ both miss, and the game is back exactly where it started, so from there Alice wins with probability p . Otherwise Bob has won. Hence

$$p = \frac{1}{6} + \frac{25}{36}p \quad \implies \quad p = \boxed{\frac{6}{11}} \approx 0.545.$$

As a check, Alice wins in round k with probability $(25/36)^{k-1} \cdot \frac{1}{6}$, and $\sum_{k \geq 1} (25/36)^{k-1} \cdot \frac{1}{6} = \frac{1/6}{1-25/36} = \frac{6}{11}$. Going first is worth about 4.5 percentage points. The series worked here, but the first-step equation keeps working when the series becomes hopeless.

Example 6.6 (Gambler's ruin). Alice has \$1 and Bob has \$2. They flip a fair coin repeatedly, and each flip moves \$1 from the loser to the winner, until one of them has all \$3. What is the probability that Alice ends up with everything?

Let p_i be the probability that Alice wins everything *given that Alice currently has \$i*, so $p_0 = 0$ and $p_3 = 1$. Conditioning on the next flip,

$$p_1 = \frac{1}{2}p_2 + \frac{1}{2}p_0 = \frac{1}{2}p_2, \quad p_2 = \frac{1}{2}p_3 + \frac{1}{2}p_1 = \frac{1}{2} + \frac{1}{2}p_1.$$

Substituting the second into the first, $p_1 = \frac{1}{4} + \frac{1}{4}p_1$, so $p_1 = \boxed{\frac{1}{3}}$ and $p_2 = \frac{2}{3}$.

Remark 6.7. With a fair coin and \$ N in total, a player holding \$ i wins everything with probability i/N — simply their share of the money. The intuition is that a fair game cannot change your expected fortune, so $i = Np_i + 0 \cdot (1 - p_i)$; turning that into a proof takes more theory than we have, but the system of equations above is rigorous and works for any N . With a biased coin the answer changes dramatically, and that is where the theory of random walks begins.

6.3 Bayes' theorem

The law of total probability computes $\mathbb{P}(B)$ from the pieces. Bayes' theorem runs the arrow backwards: given that B happened, which piece were we in?

Theorem 6.8 (Bayes' theorem). *Let $\{A_i\}$ be a partition of S with $\mathbb{P}(A_i) > 0$, and let $\mathbb{P}(B) > 0$. Then*

$$\mathbb{P}(A_i | B) = \frac{\mathbb{P}(A_i) \mathbb{P}(B | A_i)}{\sum_j \mathbb{P}(A_j) \mathbb{P}(B | A_j)}$$

Proof. By the definition of conditional probability and the multiplication rule,

$$\mathbb{P}(A_i | B) = \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A_i) \mathbb{P}(B | A_i)}{\mathbb{P}(B)},$$

and expand the denominator with the law of total probability. □

6.3.1 Vocabulary

$\mathbb{P}(A_i)$	prior	what you believed before the evidence
$\mathbb{P}(B \mid A_i)$	likelihood	how well cause i explains the evidence
$\mathbb{P}(A_i \mid B)$	posterior	what you believe after the evidence

Example 6.9 (Lost umbrella, continued). With $\mathbb{P}(B) = 0.59$ from above,

$$\mathbb{P}(A_1 \mid B) = \frac{0.40}{0.59} \approx 0.678, \quad \mathbb{P}(A_2 \mid B) = \frac{0.18}{0.59} \approx 0.305, \quad \mathbb{P}(A_3 \mid B) = \frac{0.01}{0.59} \approx 0.017.$$

The bus started at 20% and dropped to under 2%: recovering it at all is strong evidence it was never on the bus.

Example 6.10 (n heads in a row). A coin is two-headed with probability $\mathbb{P}(A) = 0.05$, and fair with probability $\mathbb{P}(A^c) = 0.95$. You toss it n times and get all heads; call that B . Then $\mathbb{P}(B \mid A) = 1$ and $\mathbb{P}(B \mid A^c) = (1/2)^n$, so

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B \mid A)\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\mathbb{P}(A^c)} = \frac{1 \cdot 0.05}{1 \cdot 0.05 + (1/2)^n \cdot 0.95} \xrightarrow{n \rightarrow \infty} 1.$$

n	1	2	3	5	10	20
$\mathbb{P}(A \mid B)$	0.095	0.174	0.296	0.627	0.982	0.99998

Remark 6.11. Even a heavily skeptical prior is overwhelmed by enough evidence — but notice how slowly it moves at first. After three heads the two-headed hypothesis is still under 30%. Cheap evidence does not flip strong priors; that is the whole content of the disease-testing example we set up last lecture and finish next time.